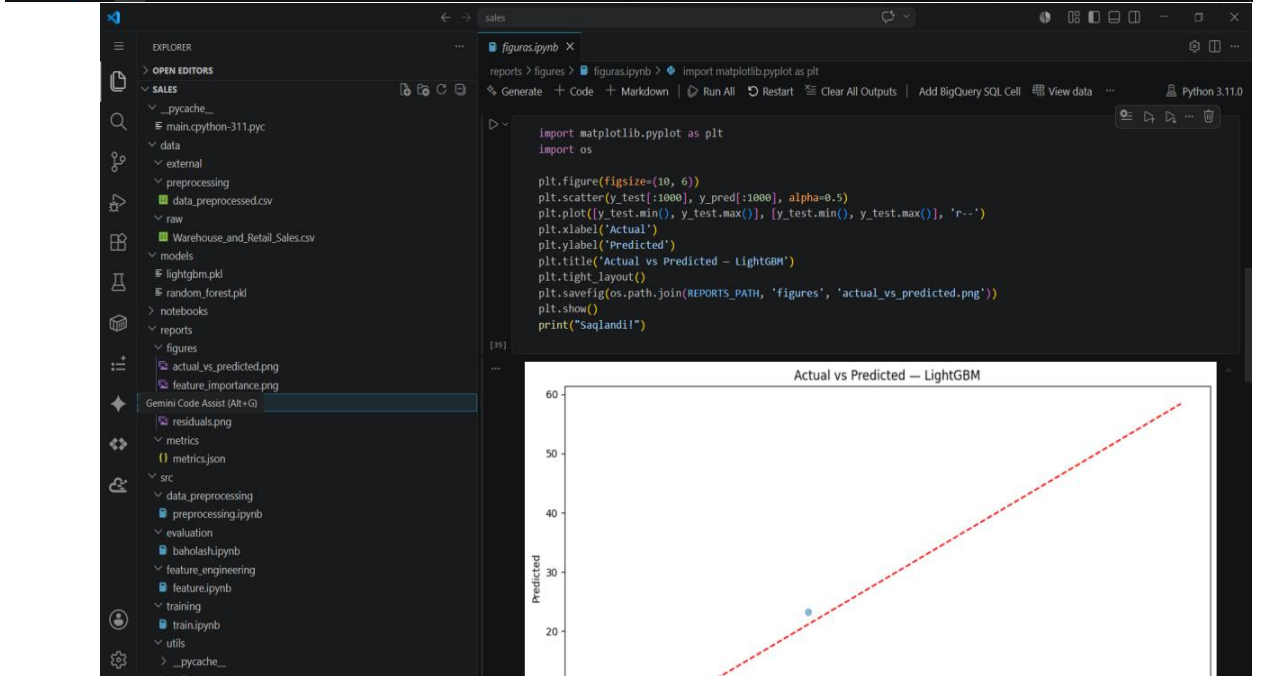
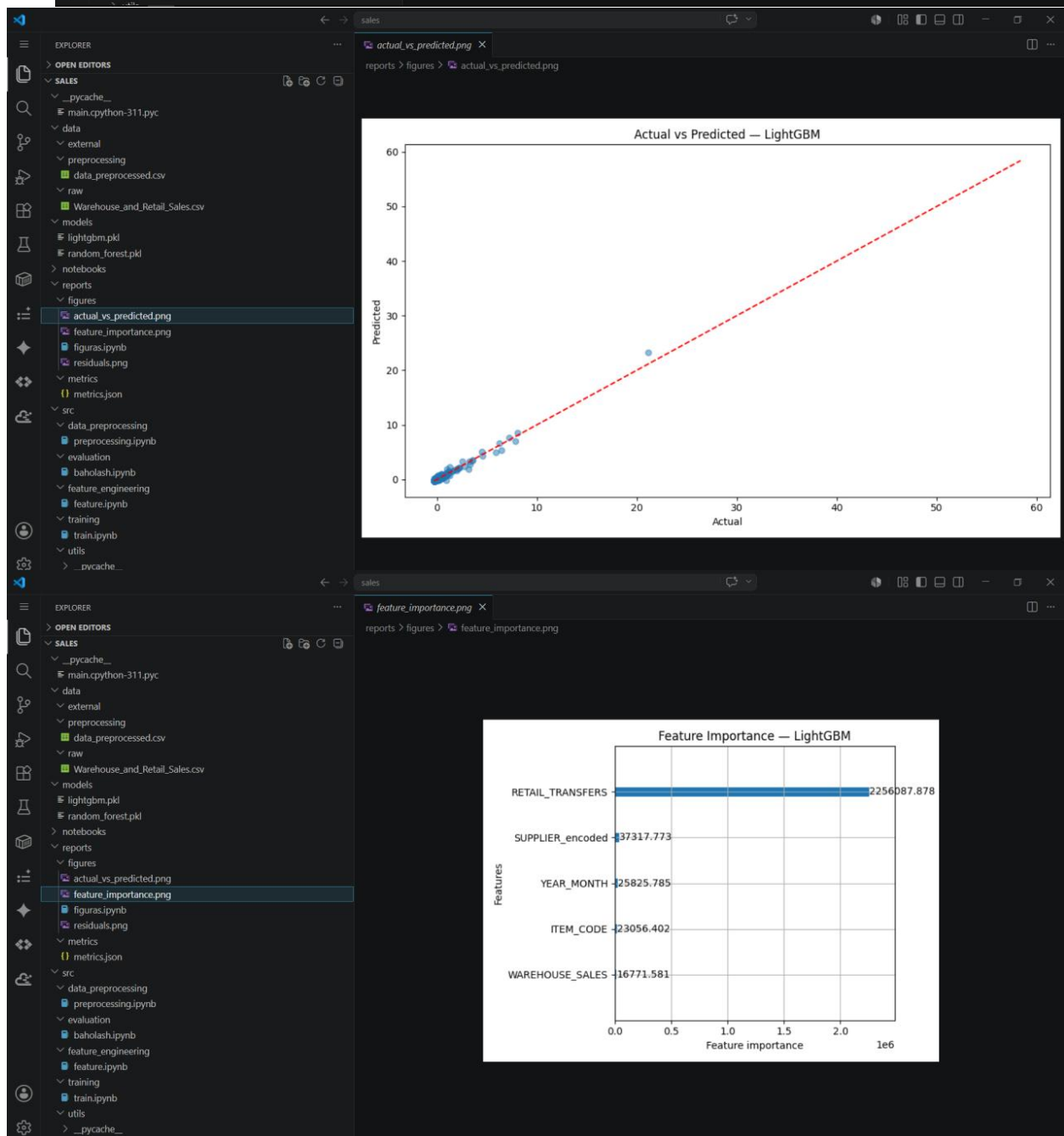
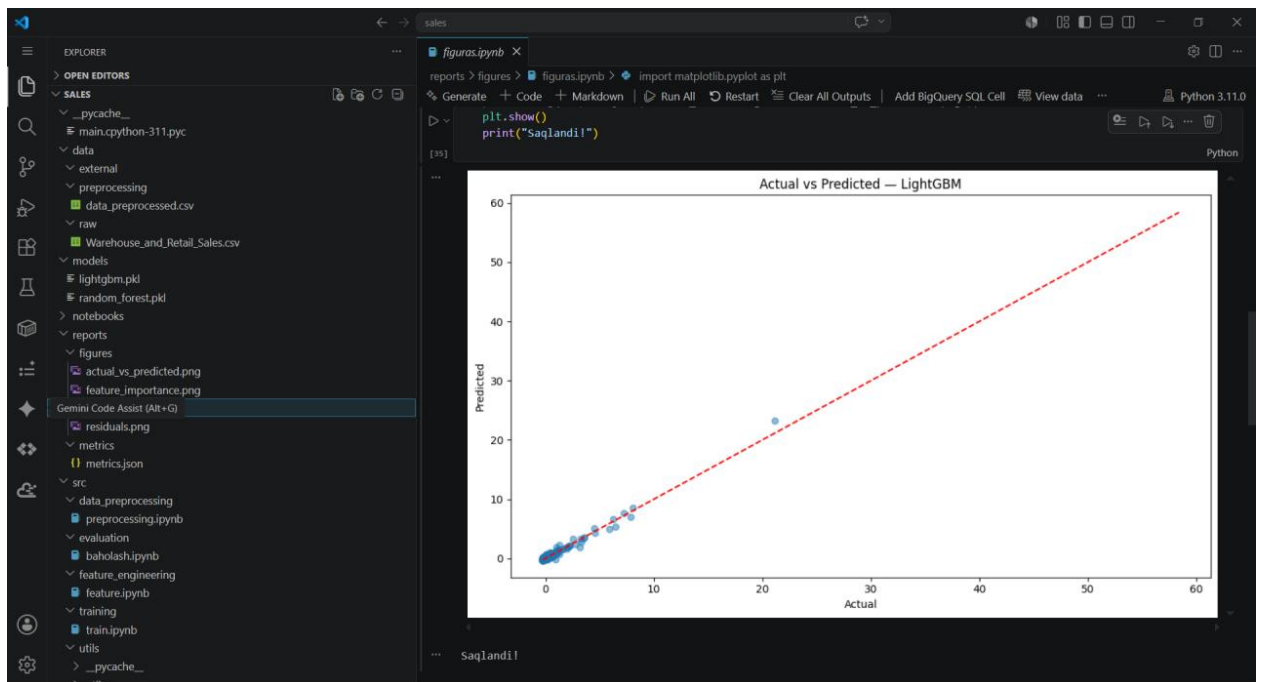
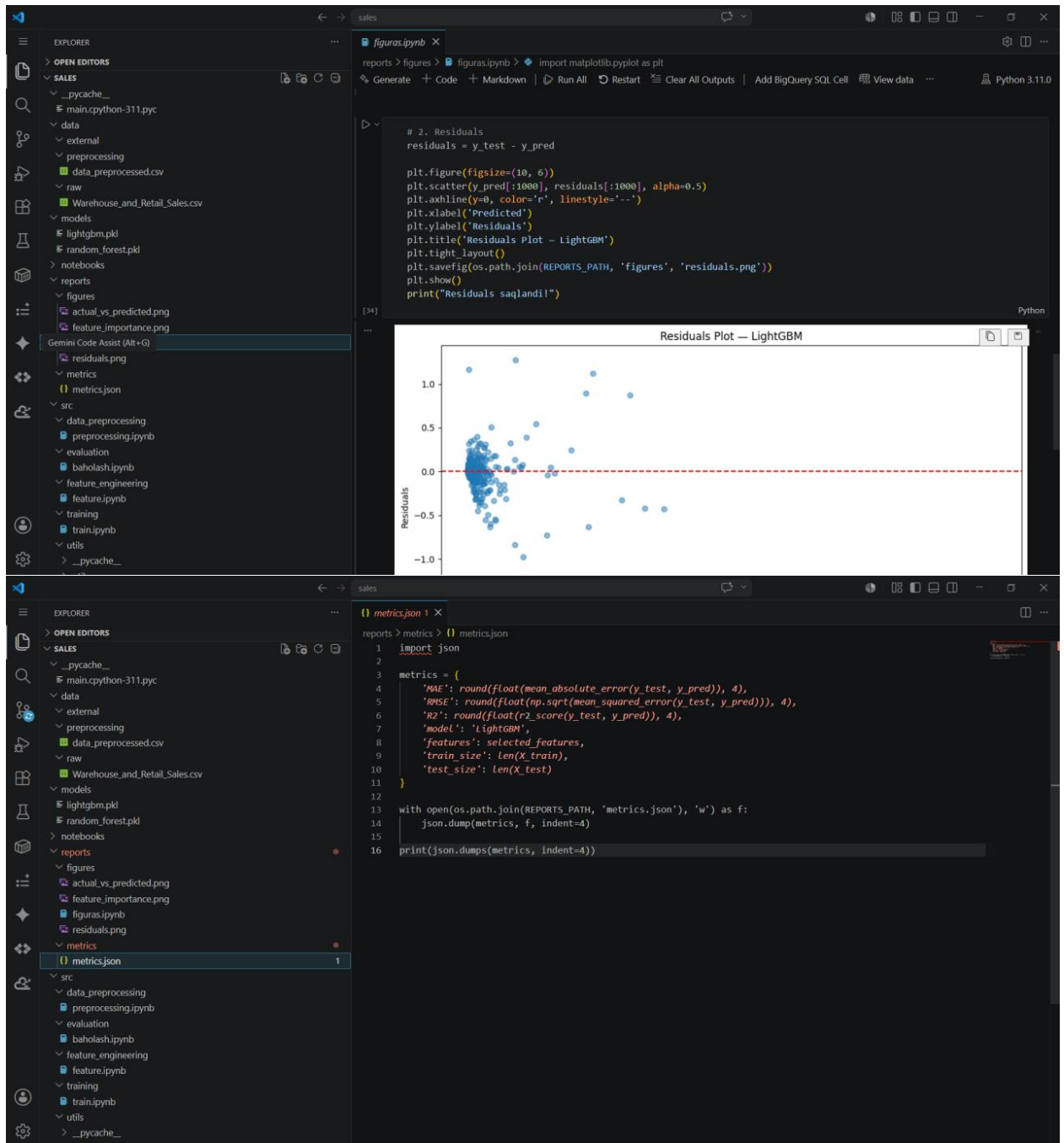


```
data > preprocessing > data_preprocessed.csv X
1 YEAR,MONTH,ITEM CODE,RETAIL SALES,RETAIL TRANSFERS,WAREHOUSE SALES,type_BEER,type_DUNNAGE,type_KEGS,type_
2 2020,1,3,-0.22668278816962525,-0.22940208889531527,-0.09320956002577693,0,0,0,0,0,0,1,-0.495263568154
3 2020,1,8,-0.22668278816962525,-0.19633018491039153,-0.08520688367909351,0,0,0,0,0,0,0,1,-0.612703980789
4 2020,1,11,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.38611767641
5 2020,1,13,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.62203921356
6 2020,1,20,-0.2002192617257796,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.608480951392
7 2020,1,46,-0.13761043087082775,-0.22940208889531527,-0.0772042073324101,0,0,0,0,0,0,0,1,-0.483692191197
8 2020,1,53,-0.2002192617257796,-0.19633018491039153,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.2877445692686
9 2020,1,59,-0.22668278816962525,-0.22940208889531527,-0.09320956002577693,0,0,0,0,0,0,0,1,-0.192249874257
10 2020,1,80,-0.01981545650249046,-0.09711447295562027,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.818218191316
11 2020,1,86,-0.21603283240563861,-0.19633018491039153,-0.09320956002577693,0,0,0,0,0,0,0,1,-0.62872179786
12 2020,1,92,-0.17181937968848188,-0.19633018491039153,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.202762190705
13 2020,1,95,-0.19376474308093922,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.54920034738
14 2020,1,96,-0.20473742477716791,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.54920034738
15 2020,1,97,-0.21571010647339658,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.54920034738
16 2020,1,102,-0.22668278816962525,-0.22940208889531527,-0.09320956002577693,0,0,0,0,0,0,0,1,-0.5952156082
17 2020,1,115,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.5853132546
18 2020,1,120,-0.21571010647339658,-0.19633018491039153,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.4952635681
19 2020,1,132,-0.2221517325375295,-0.22940208889531527,-0.09320956002577693,0,0,0,0,0,0,0,1,-0.5442681908
20 2020,1,137,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.4952635681
21 2020,1,140,-0.22668278816962525,-0.22940208889531527,-0.08920822185243522,0,0,0,0,0,0,0,1,-0.5442681908
22 2020,1,151,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.5442681908
23 2020,1,152,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.5442681908
24 2020,1,156,-0.18782459764224414,0.16216925428618192,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.818218191316
25 2020,1,157,-0.13380207814471104,0.13438885493884598,-0.06520019281238497,0,0,0,0,0,0,0,1,-0.7470759925838
26 2020,1,159,-0.22345552884728057,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.35139478724
27 2020,1,163,-0.18634204663937276,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.81665492785
28 2020,1,164,-0.2163555833788062,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.5952156082
29 2020,1,167,-0.22668278816962525,-0.22940208889531527,-0.0772042073324101,0,0,0,0,0,0,0,1,-0.5952156082
30 2020,1,176,-0.18782459764224414,0.16216925428618192,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.818218191316
31 2020,1,178,-0.16504213511139945,-0.09711447295562027,-0.4869844751087708,0,0,0,0,0,0,0,1,-0.624121013041
32 2020,1,179,-0.2102376562528225,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.5853132546
33 2020,1,185,-0.22668278816962525,-0.22940208889531527,-0.09721089819911864,0,0,0,0,0,0,0,1,-0.6342192630
34 2020,1,190,-0.19925108392905358,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.38611767641
35 2020,1,192,-0.22668278816962525,-0.22940208889531527,-0.06520019281238497,0,0,0,0,0,0,0,1,-0.4412092311
36 2020,1,193,-0.2241009087116891,-0.22940208889531527,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.511717491429
37 2020,1,197,-0.2102376562528225,-0.13018637694054402,-0.10121223637246035,0,0,0,0,0,0,0,1,-0.3842368898
```

```
Warehouse_and_Retail_Sales.csv X
1 YEAR,MONTH,SUPPLIER,ITEM CODE,ITEM DESCRIPTION,ITEM TYPE,RETAIL SALES,RETAIL TRANSFERS,WAREHOUSE SALES
2 2020,1,REPUBLIC NATIONAL DISTRIBUTING CO,100009,BOOTLEG RED - 750ML,WINE,0.00,0.00,2.00
3 2020,1,PWSN INC,100024,MOMENT DE PLAISIR - 750ML,WINE,0.00,1.00,4.00
4 2020,1,RELIABLE CHURCHILL LLP,1001,S SMITH ORGANIC PEAR CIDER - 18.70Z,BEER,0.00,0.00,1.00
5 2020,1,LANTERNA DISTRIBUTORS INC,100145,SCHLINK HAUS KABINETT - 750ML,WINE,0.00,0.00,1.00
6 2020,1,DIONYSOS IMPORTS INC,100293,SANTORINI GAVALA WHITE - 750ML,WINE,0.82,0.00,0.00
7 2020,1,KYSELA PERE ET FILS LTD,100641,CORTENOVA VENETO P/GRIG - 750ML,WINE,2.76,0.00,6.00
8 2020,1,SANTA MARGHERITA USA INC,100749,SANTA MARGHERITA P/GRIG ALTO - 375ML,WINE,0.00,1.00,1.00
9 2020,1,BROWN-FORMAN BEVERAGES WORLDWIDE,1008,JACK DANIELS COUNTRY COCKTAIL SOUTHERN PEACH - 10.0Z-NR,BEER
10 2020,1,JIM BEAM BRANDS CO,10103,KNOB CREEK BOURBON 9YR - 100P - 375ML,LIQUOR,6.41,4.00,0.00
11 2020,1,INTERNATIONAL CELLARS LLC,101117,KSARA CAB - 750ML,WINE,0.33,1.00,2.00
12 2020,1,HEAVEN HILL DISTILLERIES INC,10120,J W DANT BOURBON 100P - 1.75L,LIQUOR,1.70,1.00,0.00
13 2020,1,BACCHUS IMPORTERS LTD,10123,NOTEWORTHY SMALL BATCH BOURBON - 750ML,LIQUOR,1.02,0.00,0.00
14 2020,1,BACCHUS IMPORTERS LTD,10124,NOTEWORTHY SMALL BATCH RYE 750ML,LIQUOR,0.68,0.00,0.00
15 2020,1,BACCHUS IMPORTERS LTD,10125,NOTEWORTHY SMALL BATCH HONEY BBN 750,LIQUOR,0.34,0.00,0.00
16 2020,1,MONSIEUR TOUTON SELECTION,101346,ALSACE WILLIAM GEW - 750ML,WINE,0.00,0.00,2.00
17 2020,1,"THE COUNTRY VINTNER, LLC DBA WINEBOW",101486,POLIZIANO ROSSO MONTEPUL - 750ML,WINE,0.00,0.00,1.00
18 2020,1,"THE COUNTRY VINTNER, LLC DBA WINEBOW",101532,HATSUMAGO SAKI JUN MAT SHU - 720ML,WINE,0.34,1.00,1.1
19 2020,1,ROYAL WINE CORP,101664,RAMON CORDOVA RIOJA - 750ML,WINE,0.16,0.00,2.00
20 2020,1,REPUBLIC NATIONAL DISTRIBUTING CO,101702,MANISCHEWITZ CREAM RED CONCORD - 1.5L,WINE,0.00,0.00,1.00
21 2020,1,ROYAL WINE CORP,101753,BARKAN CLASSIC PET SYR - 750ML,WINE,0.00,0.00,3.00
22 2020,1,ROYAL WINE CORP,101915,WEINSTOCK CAB - 750ML,WINE,0.00,0.00,1.00
23 2020,1,ROYAL WINE CORP,101923,WEINSTOCK CHARD - 750ML,WINE,0.00,0.00,1.00
24 2020,1,JIM BEAM BRANDS CO,10197,KNOB CREEK BOURBON 9YR - 100P - 1.75L,LIQUOR,12.72,11.84,0.00
25 2020,1,STE MICHELLE WINE ESTATES,101974,CH ST MICH P/GRIS - 750ML,WINE,11.17,11.00,9.00
26 2020,1,Default,102,TWO BOTTLE WINE TOTE,STR SUPPLIES,0.10,0.00,0.00
27 2020,1,SAZERAC CO,102032,FUKI SAKI - 750ML,WINE,1.25,0.00,1.00
28 2020,1,MONSIEUR TOUTON SELECTION,102083,CH DE LA CHESNAIE MUSCADET - 750ML,WINE,0.32,0.00,0.00
29 2020,1,MONSIEUR TOUTON SELECTION,102148,MONSIEUR TOUTON S/BLC - 750ML,WINE,0.00,0.00,6.00
30 2020,1,JIM BEAM BRANDS CO,10243,OLD GRAND DAD BOURBON 100P
31 2020,1,SALVETO IMPORTS LLC,10245,CRICOVA ORIGINAL WH - 750
32 2020,1,"THE COUNTRY VINTNER, LLC DBA WINEBOW",10246,HILLROC
33 2020,1,HUB WINE CORPORATION,10264,90+ CELLARS LOT 113 BIG F
34 2020,1,RELIABLE CHURCHILL LLP,10269,FEN BOURBON - 750ML,L
35 2020,1,ALLAGASH BREWING COMPANY,10271,ALLAGASH SAISON 6/4 F
36 2020,1,CAMPARI AMERICA LLC,10272,WILD TURKEY AMERICAN HONEY
37 2020,1,PERNOD RICARD USA LLC,10276,ABERLOUR 18 YR - 750ML,t
38 2020,1,CAMPARI AMERICA LLC,10276,WILD TURKEY 10Y - 750ML,LIQUOR,1.14,0.00,0.00
```







The image displays two sequential screenshots of a Jupyter Notebook titled 'baholash.ipynb' within the Visual Studio Code editor. The Explorer sidebar on the left shows a project structure for a 'sales' dataset, including files for data preprocessing, model training, and evaluation.

Top Screenshot: The notebook cell contains code for loading the dataset, selecting features, training a Random Forest model, and evaluating its performance. The output shows the following metrics:

```
MAE: 0.03
RMSE: 0.13
R²: 0.9835
```

Bottom Screenshot: This screenshot shows the next steps in the notebook, including the import of `train_test_split` from `sklearn.model_selection`, the splitting of the data into training and testing sets, and the training of a LightGBM model. The output for the LightGBM model training is partially visible:

```
Test natijalar:
MAE: 0.05
RMSE: 0.21
R²: 0.9563
```

```
import pandas as pd
import numpy as np
import joblib
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from utils.config import DATA_PROCESSED_PATH, TARGET, MODELS_PATH

# Data yuklash
df = pd.read_csv(DATA_PROCESSED_PATH)
df['YEAR_MONTH'] = df['YEAR'] * 100 + df['MONTH']

selected_features = ['RETAIL TRANSFERS', 'ITEM CODE', 'YEAR_MONTH', 'SUPPLIER_encoded', 'WAREHOUSE SALES']

X = df[selected_features]
y = df[TARGET]

# Modelni yuklash
model = joblib.load(os.path.join(MODELS_PATH, 'random_forest.pkl'))

# Bashorat
y_pred = model.predict(X)

print(f"MAE: {mean_absolute_error(y, y_pred):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y, y_pred)):.2f}")
print(f"R²: {r2_score(y, y_pred):.4f}")

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

y_pred_test = model.predict(X_test)

print("Test natijalar:")
print(f"MAE: {mean_absolute_error(y_test, y_pred_test):.2f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred_test)):.2f}")
print(f"R²: {r2_score(y_test, y_pred_test):.4f}")

import lightgbm as lgb

model_lgb = lgb.LGBMRegressor(
    n_estimators=500,
    learning_rate=0.05,
    max_depth=8,
    random_state=42,
    n_jobs=-1
)

model_lgb.fit(X_train, y_train)
print("LightGBM natijalar:")
```

Explorer view showing project structure:

- SALES
 - __pycache__
 - main.python-311.pyc
 - data
 - external
 - preprocessing
 - data_preprocessed.csv
 - raw
 - Warehouse_and_Retail_Sales.csv
 - models
 - lightgbm.pkl
 - random_forest.pkl
 - notebooks
 - reports
 - figures
 - actual_vs_predicted.png
 - feature_importance.png
 - figuras.ipynb
 - residuals.png
 - metrics
 - metrics.json
 - src
 - data_preprocessing
 - preprocessing.ipynb
 - evaluation
 - baholash.ipynb
 - feature_engineering
 - feature.ipynb
 - training
 - train.ipynb
 - utils
 - __pycache__

Feature importance results:

Feature	Importance
RETAIL TRANSFERS	0.9285
ITEM CODE	0.0358
YEAR_MONTH	0.0117
SUPPLIER_encoded	0.0074
WAREHOUSE SALES	0.0070
MONTH	0.0045
SEASON	0.0024
YEAR	0.0008
type_STR_SUPPLIES	0.0006
type_NON-ALCOHOL	0.0004
type_LIQUOR	0.0003

Tanlangan featurelar (5 ta): ['RETAIL TRANSFERS', 'ITEM CODE', 'YEAR_MONTH', 'SUPPLIER_encoded', 'WAREHOUSE SALES']

The image shows a Jupyter Notebook titled 'train.ipynb' in a VS Code editor. The notebook is part of a project named 'sales' located in the 'src' directory. The file explorer on the left shows the project structure, including files like 'main.cpython-311.pyc', 'data_preprocessed.csv', 'Warehouse_and_Retail_Sales.csv', 'lightgbm.pkl', 'random_forest.pkl', 'figures', 'actual_vs_predicted.png', 'feature_importance.png', 'metrics.json', 'preprocessing.ipynb', 'evaluation', 'baholash.ipynb', 'feature_engineering', 'feature.ipynb', 'training', 'train.ipynb', and 'utils'.

The notebook contains the following code cells:

```
[1]: import sys
import os
sys.path.append(os.path.abspath('../src'))
```

```
[2]: import pandas as pd
from utils.config import DATA_PROCESSED_PATH, TARGET

selected_features = ['RETAIL TRANSFERS', 'ITEM CODE', 'YEAR_MONTH', 'SUPPLIER_encoded', 'WAREHOUSE SALES']

df = pd.read_csv(DATA_PROCESSED_PATH)

# YEAR_MONTH yaratish
df['YEAR_MONTH'] = df['YEAR'] * 100 + df['MONTH']

X = df[selected_features]
y = df[TARGET]

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print(f"Train: {X_train.shape}, Test: {X_test.shape}")
```

Output of cell [2]:

```
Train: (246116, 5), Test: (61529, 5)
```

```
[3]: from sklearn.ensemble import RandomForestRegressor
```

```
[4]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print(f"Train: {X_train.shape}, Test: {X_test.shape}")
```

Output of cell [4]:

```
Train: (246116, 5), Test: (61529, 5)
```

```
[5]: from sklearn.ensemble import RandomForestRegressor
from utils.config import RANDOM_STATE, MODELS_PATH
import joblib

model = RandomForestRegressor(
    n_estimators=100,
    random_state=RANDOM_STATE,
    n_jobs=-1
)

model.fit(X_train, y_train)
print("Model o'rgatildi!")

# Modelni saqlash
os.makedirs(MODELS_PATH, exist_ok=True)
joblib.dump(model, os.path.join(MODELS_PATH, 'random_forest.pkl'))
print("Model saqlandi!")
```

Output of cell [5]:

```
Model o'rgatildi!
Model saqlandi!
```

